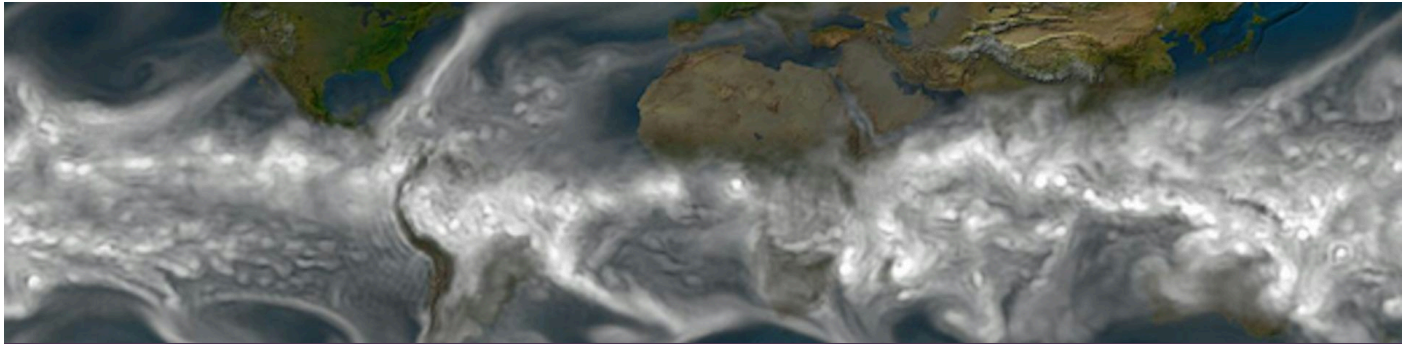




Unpacking Climate Projections: Physics, Models, and the Future

Abstract

In 2017, a U.S. congressional committee chairman claimed that predicting climate far into the future is impossible and not credible science. Climate scientists Kevin Trenberth and Reto Knutti responded by explaining how climate models work, why they rely on the laws of physics, and why projections about the future are sound, transparent, and useful for decisions.

Introduction

At a March 29, 2017 hearing of the House Committee on Science, Space and Technology, Chairman Lamar Smith argued that the scientific method depends on reproducibility and that long-range climate prediction amounts to wild guesses. At the same time, the head of the Environmental Protection Agency said that measuring human influence on climate with precision was very challenging. Trenberth and Knutti called these statements misguided. Science, they explained, is not a matter of opinion: it is based on evidence and physical principles, often combined in a model. Because we cannot run experiments on a second Earth, climate scientists build a virtual planet inside a computer.

How Do Scientists Study This?



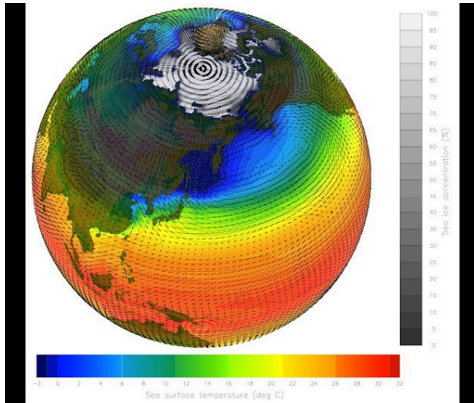
FROM THE ARTICLE

House Science Committee Chairman Lamar Smith, who questioned climate projections in 2017. Image: NASA, CC BY-NC.

Climate science starts with enormous amounts of data: millions of daily observations, mostly collected for weather forecasting, plus satellite measurements. Scientists carefully adjust for known problems, such as moved weather stations or new instruments, using overlapping observations. Climate models then represent the physical laws of nature as mathematical equations, solved on a three-dimensional grid covering the globe. The same kind of models are used every day for weather forecasting. Because weather is chaotic, scientists run many simulations with small changes in starting conditions, called ensembles, to identify robust features. For climate, the focus is on averages, patterns, and variability rather than any single day's weather.

Explore the Evidence

FROM THE ARTICLE



One month from a 20th-century simulation by NCAR's Community Climate System Model, showing winds, sea surface temperatures, and sea ice. Image: NCAR, CC BY-NC-ND.



Satellites provide a main source of data about the Earth system. Image: NASA, CC BY.

How climate projections are made

1 Observations

Weather stations, ships, buoys, satellites

2 Physics

Laws of nature as equations

3 Ensembles

Many runs reveal robust patterns

4 What-if scenarios

Projections with uncertainty ranges

Gather observations

Millions of daily weather and satellite measurements are checked and assembled.

Build a virtual Earth

Physical laws become equations solved on a 3D grid around the globe.

Run ensembles

Many runs with slightly different starts reveal which results are robust.

What Did They Discover?

The authors explained that climate models realistically simulate the annual cycle, past climates, 20th-century change, and natural variability. Detailed weather cannot be forecast more than about two weeks ahead, but long-term climate can be projected, much as planetary orbits can be predicted for millions of years, because the laws of physics 500 years from now will be the same as today. Because human choices are unpredictable, scientists make projections based on what-if scenarios, not predictions, and attach likelihoods and uncertainty ranges. The science is transparent and reproducible: data and models are mostly public, and running the same code twice gives the same answer. Reviews show early model projections of temperature and precipitation were remarkably accurate.

Models by the numbers

Millions

of daily observations feed climate science

~2 wks

limit for detailed day-to-day weather forecasts

500 yrs

from now, physics will still be the same

The authors compare climate risk to boarding a plane with a cracked wing: we do not need certainty of disaster to justify taking precautions.

Takeaways

Physics, not guesswork

Climate models are built on the laws of physics, which do not change over time.

Projections, not predictions

Models answer what-if questions about different human choices, with stated uncertainty.

Open and tested

Data and models are public, reproducible, and early projections proved accurate.

Key Terms

Climate model

A computer representation of Earth's climate system built from physical laws expressed as mathematical equations.

Projection

An estimate of future climate based on a specific what-if scenario of human activity, rather than a certain prediction.

Ensemble

A set of many model runs with slightly different starting conditions, used to find reliable patterns.

Reproducibility

The ability of other scientists to repeat a study and get the same results.

Chaos

Sensitivity of a system to tiny changes in starting conditions, which limits detailed weather forecasts to about two weeks.

Words to Know · Unpacking Climate Projections

Climate model

Projection

Ensemble

Reproducibility

Chaos

Activity

Weather vs. Climate Dice

Students use dice to see why single outcomes are unpredictable but long-run patterns are reliable, like weather and climate.

Materials: Two dice per group; Tally sheet; Graph paper

1. Predict the result of a single roll of two dice. Roll once. Were you right?
2. Roll 50 times and tally the sums. Graph the results.
3. Compare your graph with other groups. Which sums are most common for everyone?
4. Now change the rules to 'load' the dice by adding 1 to every roll. How does the pattern shift?
5. Explain how this models the difference between weather forecasts and climate projections, and how adding greenhouse gases is like loading the dice.

References

Trenberth, K., and Knutti, R. (2017). Yes, we can do 'sound' climate science even though it's projecting the future. The Conversation. <https://doi.org/10.64628/AAI.scydsux5h>